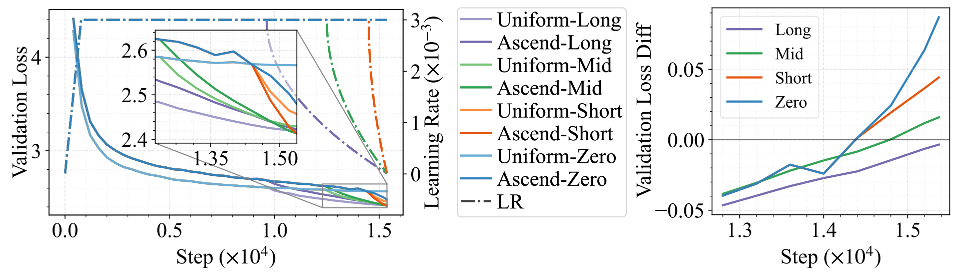
Learning Rate Schedules. The learning rate (LR) schedule is a crucial hyperparameter in model pretraining. Traditional LR schedules like cosine decay are the most widely used in LLM pretraining (Loshchilov & Hutter, 2017; Dubey et al., 2024; Li et al., 2024). More recent schedules like Warmup-Stable-Decay (WSD) have demonstrated strong performance and are suitable for mid-training resumption paradigms (Hu et al., 2024; Hägele et al., 2024). Loss curve scaling laws suggest that they possess an optimal shape for a given peak LR (Tissue et al., 2024; Luo et al., 2025; Li et al., 2025a). A prevailing finding for training with standard uniform data ordering is that an optimal or near-optimal LR schedule should decay to a value close to zero (Li et al., 2025b; OLMo et al., 2025; Li et al., 2024). Our work challenges this convention in the context of curriculum-based pretraining. We find that for a data curriculum, such an aggressive decay has negative effects on the data schedule. Instead, a moderate decay that maintains a higher learning rate during the high-quality data phase proves superior, especially when using model averaging. Unless otherwise specified, we discuss the LR schedules with a warmup phase. For example, a constant LR schedule refers to a linear warmup followed by a constant LR.

Model Averaging. Model averaging, which combines parameters from multiple checkpoints to produce a final, improved model, is a well-established technique for improving generalization (Izmailov et al., 2018; Jin et al., 2023; Yang et al., 2024b). It has been successfully applied in LLM pretraining to accelerate convergence and boost performance, and was reportedly used in training prominent models like Llama 3 (Kaddour, 2022; Li et al., 2025c; Dubey et al., 2024). Prior work has explored combining model averaging with decay-free LR schedules for training on uniformly ordered data and in the mid-training setting (Li et al., 2025c; Tian et al., 2025). However, Li et al. (2025c) finds that the performance of weight averaging is merely comparable to standard LR decay, and Tian et al. (2025) attempts to improve the results through a weighting strategy derived from the LR schedule (Discussed in Sections A.2 and 3). Unlike this prior work, which focused on uniform data ordering, our research is the first to investigate the interaction between model averaging and a quality-based data curriculum. We find that these two strategies have a strong synergistic relationship, as model averaging provides the necessary training stability to learn from high-quality data with a high, moderately decaying learning rate.

3 Preliminary

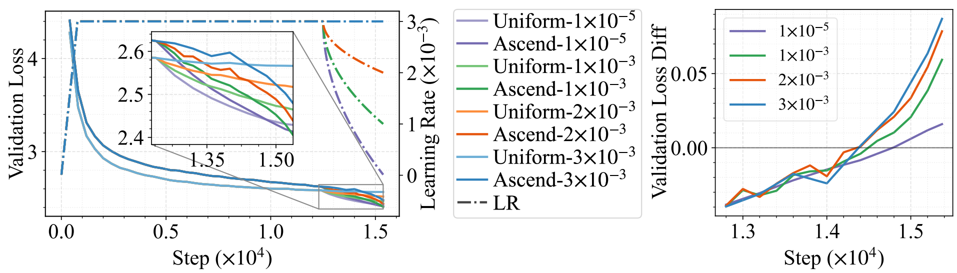
Learning Rate Schedule. We consider two primary types of learning rate (LR) schedules in addition to a constant LR schedule. The commonly used cosine schedule defines the LR at step t as $\eta(t) = \eta_0 \left(\frac{1+\alpha}{2} + \frac{1-\alpha}{2}\cos\left(\frac{\pi t}{T}\right)\right)$, where η_0 is the peak LR, T is the total number of training steps, and α is the ratio of the final LR to the peak LR. A more recent alternative is the Warmup-Stable-Decay (WSD) schedule, which consists of a linear warmup phase, a stable phase with a constant LR η_0, and a final decay phase. For the decay phase, we use the *1-sqrt* function, where the LR is given by $\eta(t) = \eta_0 \left(1 - \sqrt{r(t)}\right) + \eta_T \sqrt{r(t)}$, with $r(t) = \frac{t - t_{\text{decay}}}{T - t_{\text{decay}}}$ representing the progress through the decay period, which begins at step t_{decay}. Further details on our schedule choices are discussed in Section A.1.

Model Averaging. Model averaging computes a weighted average of several model checkpoints to produce a single, final model. We consider three common strategies. Suppose we have N checkpoints, $M_1, \ldots, M_N$, typically the last N checkpoints collected at fixed intervals, corresponding to steps $t_1, \ldots, t_N$. We focus on the model averaging strategies discussed in Li et al. (2025c) summarized as follows. *Simple Moving Average (SMA)* (Izmailov et al., 2018) applies a uniform weight to all these checkpoints: $M_{\text{avg}} = \frac{1}{N}\sum_{i=1}^{N} M_i$. *Exponential Moving Average (EMA)* assigns exponentially decaying weights. EMA is defined recursively as $M_{\text{avg}}^{(i)} = \alpha M_i + (1-\alpha) M_{\text{avg}}^{(i-1)}$, with the base case $M_{\text{avg}}^{(1)} = M_1$. The hyperparameter $\alpha \in (0, 1]$ controls the decay rate; a larger α assigns greater weight to the most recent checkpoint. *Weighted Moving Average (WMA)* uses a predefined set of normalized weights $w_1, \ldots, w_N$ (where $\sum w_i = 1$) to compute the final model as $M_{\text{avg}} = \sum_{i=1}^{N} w_i M_i$. Prior work (Tian et al., 2025) derives these weights from the learning rate schedule, where each weight is proportional to the drop in learning rate between checkpoints: $w_i \propto \eta(t_i) - \eta(t_{i+1})$ for $i < N$, and $w_N \propto \eta(t_N)$. The weight computation procedure for WMA is detailed in Section A.2.



(a) Validation loss for different decay steps.

(b) $\mathcal{L}_{\text{Uniform}} - \mathcal{L}_{\text{Ascend}}$



(c) Validation loss for different ending LRs.

(d) $\mathcal{L}_{\text{Uniform}} - \mathcal{L}_{\text{Ascend}}$

Figure 2: When varying the decay steps across 37%, 18%, 6% and 0% of training (*Long*, *Mid*, *Short*, *Zero*, respectively) and ending LRs ($1 \times 10^{-5}, 1 \times 10^{-3}, 2 \times 10^{-3}, 3 \times 10^{-3}$), the benefit of data curriculum diminishes with more aggressive LR decay. For each LR decay, we train 1.5B-parameter models with uniform and ascending ordering of data based on DCLM scores, and measure the difference in validation loss. As shown in (b) and (d), this difference becomes smaller with more decay steps or smaller ending LRs.

Data Scoring. Raw web data must pass through a processing pipeline before it is used for pretraining. The raw data first goes through heuristic filtering rules. Then in the model-based filtering phase, a scorer model assigns a quality score to each data sample. For example, DCLM Baseline dataset uses scores from a fasttext model (Joulin et al., 2017) measuring similarity to high-quality sources like OpenHermes 2.5 (Teknium, 2023) and top posts from the ELI5 subreddit. Another approach, PreSelect (SHUM et al., 2025), scores data based on its similarity to downstream tasks. Typically, these scores are used to filter the dataset by removing samples below a certain quality threshold. In contrast, our work does not use these scores to discard data; instead, we use these quality scores to define the data ordering for curriculum learning.

4 Learning Rate Decay Counteracts Data Curriculum

In this section, we analyze the critical yet often overlooked interaction between the learning rate (LR) schedule and the data schedule. We first explain how the learning rate acts as an implicit importance weight for each data sample. We then present empirical results to demonstrate three key points: (1) a data curriculum can yield significant benefits over a uniform data order under a constant LR schedule; (2) these benefits diminish when a conventional decaying LR schedule is applied, particularly during the final, high-quality data regime; and (3) while adjustments to the data schedule can mitigate this issue, the underlying conflict persists.

Analysis of Coupling between Learning Rate and Data Schedules. A key insight is that the learning rate schedule acts as an implicit importance weight for each training sample. The parameter update at training step t is $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \boldsymbol{g}_t$, where η_t is the learning rate. The gradient $\boldsymbol{g}_t$ can be decomposed into a signal component, $\mathbb{E}[\boldsymbol{g}_t]$, which points in the direction of steady improvement, and a noise component, ϵ_t. A decaying learning rate η_t serves a dual purpose: it reduces the noise ϵ_t to stabilize training, but it also shrinks the update step taken in the signal direction $\mathbb{E}[\boldsymbol{g}_t]$. While modern optimizers like Adam (Kingma & Ba, 2015) use more complex update rules, the learning rate remains a dominant factor in the update magnitude. This dual role of η_t creates a fundamental conflict in quality-based curricula. High-quality samples are intentionally processed at the end of training, but this is precisely when conventional LR schedules reduce η_t to its minimum. Consequently, the decaying learning rate diminishes the influence of the most valuable data, counteracting the intended benefit of the curriculum.

Experimental Settings. Our experiments are grounded in the DataComps-LM (DCLM) framework (Li et al., 2024) at the `1B-1x` scale, ensuring our findings are validated at a substantial scale.

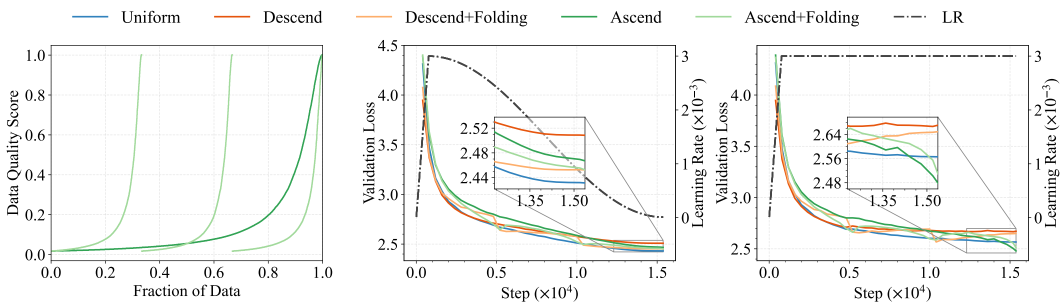


Figure 3: A stage-wise "data folding" curriculum mitigates the negative interaction observed between data ordering and learning rate (LR) decay (detailed in Section 4), but data folding can not match end-to-end sorting under a constant learning rate. **Left:** We compare simple ascending curricula (Ascend), sorted by DCLM score, against their "folding" counterparts (Ascend+Folding). The folding method involves partitioning the data into stages (three in our implementation) and performing the sort within each stage. The Descend(+Folding) curriculum is designed in reverse order. **Middle:** Under a standard cosine LR schedule, folding strategies reduce validation loss compared to simple sorting but are outperformed by a uniform data baseline. **Right:** Conversely, with a constant LR schedule where decay does not weaken the utility of high-quality data, the advantage of folding vanishes, and a simple ascending-order curriculum becomes the most effective strategy.

We adopt the Qwen2.5-1.5B model architecture (Yang et al., 2024a) and train models on a 30B token subset of the DCLM-Baseline dataset. For the data curriculum, we use DCLM's fasttext scores as our quality metric. We set the peak LR to 3×10^{-3} and the ending LR to 1×10^{-5}, aligning with optimal settings found in prior work for uniform data schedules (Li et al., 2024; Luo et al., 2025; Li et al., 2025b). We evaluate performance on a high-quality subset of the DCLM-Baseline dataset held out for validation. We also report downstream task scores by the OLMES framework (Gu et al., 2025a). Among the standard benchmark suite, we choose MMLU (Hendrycks et al., 2021), ARC (Clark et al., 2018), and CSQA (Talmor et al., 2019) as *Core* benchmarks, as recent work suggests they have a higher signal-to-noise ratio to distinguish model performance (Heineman et al., 2025). Further experimental details are provided in Section A.1.

A Data Curriculum is Highly Effective with a Constant Learning Rate. To isolate the effect of the data schedule from the LR schedule, we first conducted experiments using a constant learning rate of 3×10^{-3}. We compared three data schedules: a uniform random baseline, an ascending-order curriculum, and a reverse (descending-order) curriculum, both curricula sorted by DCLM quality scores. As shown in Figure 1(a), the ascending-order curriculum significantly outperforms the uniform baseline, achieving a much lower validation loss and faster convergence. In contrast, the reverse curriculum's validation loss trends upward, likely because the data distribution shifts progressively away from the high-quality validation set. These results clearly demonstrate that a quality-based curriculum is effective when its impact is not confounded by a decaying learning rate. Similar trends were observed using PreSelect scores (SHUM et al., 2025) (see Appendix Figure 8(a)).

The Curriculum's Advantage Diminishes with a Decaying LR Schedule. In sharp contrast to the constant LR experiments, the advantage of the data curriculum largely disappears when we employ a WSD schedule (Figure 1(b)). The performance degradation is even more pronounced with a cosine schedule, which decays throughout its entire range (Figure 1(c)). To further probe this relationship, we varied the aggressiveness of the LR decay by adjusting two WSD parameters: the number of decay steps and the final learning rate. The results in Figure 2 show a clear trend: as the decay phase becomes longer or more aggressive (i.e., a smaller ending LR), the performance benefit of the data curriculum over the uniform baseline shrinks, eventually becoming negligible. This confirms the previously overlooked effect of LR decay on the data curriculum, where LR decay undermines the contribution of high-quality data.

Exploring an Alternative Curriculum: Data Folding. We also investigated whether a different curriculum design could mitigate the coupling effect. We tested a *folding* curriculum, inspired by prior work (Dai et al., 2025; Zhang et al., 2025), where the dataset is split into several chunks and each chunk is sorted internally. This stage-wise design distributes high-quality data more evenly throughout training than the standard data curriculum. As shown in Figure 3, under a cosine schedule, the ascending with folding strategy performed better than a simple end-to-end ascending sort but still underperformed the uniform baseline. Conversely, under a constant LR schedule, the simple ascending-order curriculum proved to be the most effective strategy, outperforming the folding cur-